Presenter Notes: Continuous-Time RL for Portfolio Optimization

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1 Slide 1 — Title (~0:30; cumulative 0:30)

Introduce yourselves briefly. One sentence framing: Wu and Li (2024) extend the continuous-time mean-variance RL framework to the case where the market regime is unobservable, and this presentation covers the theory and our own application to European equity markets.

Suggested opening: "Today we're reviewing Wu and Li (2024), which asks: can you do reinforcement learning for mean-variance portfolio optimization when you can't directly observe whether you're in a bull or bear market? The answer is yes, and the structure that makes it work is surprisingly clean."

Transition: Advance immediately.

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2 Slide 2 — The Pitch (~0:20; cumulative 0:50)

Brief comic relief. The serious point embedded in the image: investors face a fundamentally adversarial environment where regimes shift without warning. Move quickly — this slide exists to set tone before the math begins.

Transition: "With that framing, here's the roadmap."

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3 Slide 3 — Outline (~0:10; cumulative 1:00)

Read the section list once at a normal speaking pace. No elaboration needed — the structure will be self-evident as you move through it.

Transition: "Before getting into motivation, here is the intellectual lineage in one picture."

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4 Slide 4 — Quick Roadmap (~0:15; cumulative 1:15)

This slide shows the three-step progression that situates the paper:

1. **Exploratory Framework** (Wang, Zariphopoulou, et al. 2020) — the general continuous-time RL scaffolding using entropy-regularized stochastic control.
2. **EMV** (Wang and Zhou 2020) — the first application of that framework to mean-variance portfolio selection, under a single GBM regime.
3. **POEMV** (Wu and Li 2024) — the extension to a regime-switching market with unobservable regime, the paper being reviewed.

Move through this quickly. Its purpose is to help the audience locate the paper in the literature before you motivate the problem.

Transition: "Starting with why this problem is worth solving."

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5 Slide 5 — Motivation (~1:10; cumulative 2:25)

Why continuous-time RL?

Financial markets trade and price continuously. The classical RL literature works in discrete time, which is a mismatch both in motivation and in tractability. The critical practical advantage of continuous time is that for linear-quadratic (LQ) problems, the optimal value function and policy take known structural forms — Gaussian policies, parabolic PDEs — that you can exploit directly. Discrete-time RL cannot offer this.

The exploration challenge is non-trivial in continuous time. You can't just randomize over a finite action set. The paper's resolution is to use distribution-valued policies and entropy regularization, covered in Slide 8.

Why regime switching?

Point to the S&P 500 chart. Bull and bear markets are not informal descriptions — they represent different data-generating processes with materially different expected returns. The EMV algorithm of Wang and Zhou (2020) fits a single GBM and ignores this structure entirely. The regime is unobservable: investors see prices $\{S_t\}$ but never observe $I_t \in \{1, 2\}$ directly. POEMV filters the current regime probability continuously from the price path.

Core point: The regime is hidden and materially important. The paper's contribution is a tractable algorithm that accounts for both facts simultaneously.

Likely question: "Why only two regimes?" Two regimes (bull/bear) is the standard in this literature (Guidolin and Timmermann 2007; Dai et al. 2010) and is empirically well-supported for long-horizon investment. More regimes would require solving higher-dimensional versions of the same PDEs — addressed in Open Problems.

Transition: "Before getting into the theory, it's worth being precise about why the classical approach fails and what RL buys you."

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6 Slide 6 — RL vs. the Classical Approach (~1:05; cumulative 3:30)

The classical failure mode.

The classical approach: estimate $(\mu_i, \lambda_i, \sigma)$ from data, plug into the analytically derived optimal control. The problem is that mean return μ is notoriously hard to estimate from financial data. Even a small error translates to large swings in optimal holdings because the optimal control is proportional to $\hat{\rho}/\sigma$, and there is no self-correcting mechanism — you estimate once and commit.

Why RL is better.

RL never uses estimated parameters to compute the control directly. Parameters are estimated only to run the Wonham filter and obtain p_t. The policy parameters (ϕ_1, ϕ_2) are learned from observed outcomes. If some configurations of the policy led to bad terminal wealth, those get corrected through actor-critic updates. This is the self-correction mechanism.

The contribution block on the slide is worth reading aloud verbatim. Wang and Zhou (2020) handled the single-regime case. Wu and Li (2024) replace the GBM with a regime-switching GBM, lose the analytical solution for f and g, but show that the structural form of the optimal policy is preserved — still Gaussian, still myopic-plus-hedging.

Core point: In the classical approach, estimation error enters the policy directly. In POEMV, estimation error enters only through the filter, and as the DAX results show, the filter is robust to moderate estimation error.

Transition: "Now the formal setup. The mean-variance problem itself."

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7 Slide 7 — The MV Portfolio Problem (~1:15; cumulative 4:45)

The constrained problem.

Over a fixed horizon $[0, T]$, minimize terminal wealth variance subject to hitting a mean target z. Via a Lagrange multiplier ω, this is equivalent to minimizing $\mathbb{E}[(X_T^u - \omega)^2] - (\omega - z)^2$, which is a mean-squared-error problem in u. The subtracted constant doesn't affect the minimization, so this is an LQ control problem — and that is why the optimal policy turns out Gaussian.

Time inconsistency.

The variance operator is not an expected value of a running reward, so the Bellman principle does not apply. A policy optimal at $t = 0$ is generally not optimal from $t > 0$. The paper focuses on the *precommitted* strategy: the investor commits at $t = 0$ and executes without re-optimizing. This is the standard approach in the continuous-time MV literature.

The ω update.

We don't know ω^* in closed form, so it's found by stochastic approximation: every $s = 10$ episodes, nudge ω to correct the gap between average terminal wealth and target z. Updates are deferred by s episodes to avoid destabilizing the policy loop while it's still actively changing.

Core point: The Lagrangian reformulation converts a constrained variance problem into an unconstrained LQ problem, which is exactly the class where Gaussian optimality holds.

Likely question: "Isn't the precommitted strategy suboptimal?" Relative to a time-consistent equilibrium strategy, yes. Extending to equilibrium strategies under partial information is one of the open problems on the final slide.

Transition: "To actually do RL in this setting, we need to formalize what exploration means in continuous time."

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8 Slide 8 — Exploratory Stochastic Control (~1:30; cumulative 6:15)

The key idea.

Replace point-action controls u_t with distribution-valued policies $\boldsymbol{\pi}_t \in \mathcal{P}(\mathcal{U})$. Instead of choosing a specific dollar amount to invest, the agent samples from a distribution over dollar amounts. This is exploration in continuous time.

The exploratory SDE.

Walk through the three terms. The first two involve $\mathbb{E}_{\boldsymbol{\pi}}[u]$ — the mean action — and are the standard drift and diffusion you'd get from a deterministic control set to the policy mean. The third term involves $\sqrt{\mathrm{Var}_{\boldsymbol{\pi}}[u]}$ and an *independent* Brownian motion $\bar{B}$. This extra noise source is introduced purely by the randomization of actions, and it is what enables the Gaussian optimality result downstream.

Entropy regularization.

Adding $-\Lambda \int_0^T \mathcal{H}(\boldsymbol{\pi}_t)\,dt$ to the objective penalizes premature collapse to a deterministic policy. The temperature Λ controls the exploration–exploitation tradeoff: $\Lambda \to 0$ recovers the original deterministic problem. For LQ problems, completing the square in the HJB equation shows the optimal exploratory policy is Gaussian. This is not an assumption — it follows from the problem structure.

Regime switching adds one complication.

In the GBM case, the wealth X_t is the only state and the covariance structure is simple. Here, the joint state is (X_t, p_t), and $\mathrm{Cov}(\Delta X^\pi, \Delta p_t) \neq 0$ because both are driven by the same innovation $\hat{B}_t$. This cross term appears in the HJB and is the source of the hedging demand in the optimal policy.

Core point: Entropy regularization is not a training trick — the optimal policy for the regularized problem is provably Gaussian, which gives a natural and parsimonious parametrization for learning.

Transition: "Now the market model and how we handle the unobservable regime."

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9 Slide 9 — Market Model & Wonham Filtering (~1:30; cumulative 7:45)

The regime-switching GBM.

The risky asset follows $dS_t = S_t(\mu_t\,dt + \sigma\,dB_t)$ where $\mu_t \in \{\mu_1, \mu_2\}$ and I_t is a continuous-time Markov chain. Regime 1 is bull ($\mu_1 > 0$), Regime 2 is bear ($\mu_2 < 0$). Volatility σ is regime-independent because it can be identified exactly from quadratic variation — it is not part of the unobservable state.

The Wonham filter.

The investor observes prices but never I_t directly. The belief state $p_t = P(I_t = 1 \mid \mathcal{F}_t^S)$ evolves according to the Wonham filter SDE on the slide. The drift term is mean-reverting toward the stationary bull probability $\lambda_2/(\lambda_1 + \lambda_2)$. The diffusion coefficient $(\rho_1 - \rho_2)p_t(1 - p_t)$ is zero at the extremes (certainty) and maximal at $p_t = 1/2$ (maximum uncertainty).

Entrance boundaries.

Both $p = 0$ and $p = 1$ are entrance boundaries: a belief process starting in $(0, 1)$ stays there forever almost surely. You never become fully certain of the regime. This is mathematically convenient — the PDEs for f and g need no boundary conditions at the endpoints.

Core point: The Wonham filter is the mechanism by which the investor continuously updates regime beliefs from observed price innovations. This filtered belief state p_t is the key additional state variable that makes POEMV different from EMV.

Likely question: "Why not use the EM algorithm for parameter estimation?" EM at daily frequency produces wildly implausible transition rates (the appendix slide shows estimates near 128 and 132, implying regime switches every two days) because the drift signal is swamped by diffusion noise. The heuristic approach of Dai et al. (2010) — segment by bull/bear periods using return thresholds, then estimate within segments — is far more stable, and σ is well-estimated by both.

Transition: "With the belief state defined, the key insight is a separation principle that makes this whole approach tractable."

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10 Slide 10 — Separation Principle (~0:30; cumulative 8:15)

The key result (Xiong and Zhou 2007).

p_t is a *sufficient statistic* for the unobservable drift μ_t. This means the partial-information problem reduces to a fully observed control problem in the augmented state (X_t, p_t). Substituting the filtered Sharpe ratio $\hat{\rho}_t$ for ρ_t, wealth dynamics become $dX_t = \sigma u_t(\hat{\rho}_t\, dt + d\hat{B}_t)$ — no hidden state remains. This yields a tractable, fully observed HJB equation.

Two important consequences.

First, the entrance-boundary property means $p_t \in (0, 1)$ strictly, so no boundary conditions are needed and standard martingale arguments apply. Second, the separation is valid for the *precommitted* strategy of the MV problem — the filtering and learning steps can be executed independently.

Core point: The separation principle converts a POMDP into a fully observed MDP by promoting p_t to a state variable. The price is needing to run the Wonham filter, which requires pre-estimated model parameters. This is the only step requiring parameter estimation; policy learning itself is model-free.

Transition: "With the state space established, we can write down what the optimal value function looks like."

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11 Slide 11 — Value Function Structure (Theorem 1) (~1:30; cumulative 9:45)

The ansatz.

The terminal condition $(x - \omega)^2 - (\omega - z)^2$ is quadratic in x, motivating the separating ansatz $v = (x - \omega)^2/f(t, p) + g(t, p) - (\omega - z)^2$. Plugging into the HJB and matching terms decouples the problem: f and g each satisfy a linear parabolic PDE in (t, p) alone. Two scalar functions of two variables replace one nonlinear PDE in three variables — that decoupling is the core theoretical contribution.

Key properties (Theorem 1).

f is bounded, decreasing in t, with f_p and f_{pp} uniformly bounded. The PDEs are degenerate at $p \in \{0, 1\}$ because the second-order coefficient $(\rho_1 - \rho_2)^2 p^2 (1 - p)^2$ vanishes there — but entrance boundaries mean no boundary conditions are needed. Well-posedness is established via a comparison principle using Itˆo's formula on a stopped diffusion.

The two PDEs are solved sequentially: f first (it doesn't depend on g), then g uses f as input. Both are solved numerically by finite differences — there is no closed form once market parameters are unknown.

f vs. g roles.

f encodes exploration variance: the policy variance is $f(t, p)\Lambda/(2\sigma^2)$. g encodes the entropy bonus accumulated under the regularized objective. When $\Lambda = 0$, $g \equiv 0$ and the problem reduces to the classical deterministic case.

Core point: The ansatz separates x from p, converting an intractable nonlinear HJB into two tractable linear PDEs. The entrance-boundary structure means no boundary conditions are needed despite the degeneracy.

Transition: "With f and g in hand, we can characterize the optimal policy explicitly."

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12 Slide 12 — Optimal Policy: Myopic–Hedging Decomposition (Theorem 2) (~1:45; cumulative 11:30)

The Gaussian policy.

Minimizing over $\boldsymbol{\pi}$ in the HJB yields a Gaussian policy with variance $f(t,p)\Lambda/(2\sigma^2)$. The mean decomposes into two terms worth dwelling on.

The myopic term $-(x-\omega)\hat{\rho}_t/\sigma$ is what you would invest under a single-regime model with the filtered Sharpe ratio $\hat{\rho}_t$. It scales position by the wealth deviation $(x-\omega)$ and the current belief about the market.

The hedging term $+(x-\omega)(\rho_1-\rho_2)p(1-p)(f_p/f)/\sigma$ accounts for regime transition risk. When $\rho_1 > \rho_2$ and $f_p > 0$ (higher belief in bull means better prospects), the hedging term *reduces* the position relative to the myopic allocation, because the investor hedges against the possibility of switching out of the bull regime. This hedging demand is largest at $p_t \approx 1/2$ (maximum uncertainty) and vanishes as beliefs concentrate near 0 or 1.

Recovery of EMV. Setting $\rho_1 = \rho_2$ makes $f_p = 0$ and collapses the hedging term entirely, recovering the Wang-Zhou (2020) EMV policy. EMV is a special case of POEMV with no regime structure.

Policy variance decays toward T. Since $f(t,p)$ is decreasing in t, exploration naturally winds down as the horizon approaches — the continuous-time analog of decaying ε-greedy.

Core point: The hedging demand is the exclusive contribution of regime switching. At maximum uncertainty, POEMV pulls back from the risky asset relative to the myopic benchmark; at high confidence in either regime, it converges to the myopic allocation.

Suggested phrasing for hedging: "Think of the hedging demand as uncertainty insurance. When you're most unsure about the regime, you reduce your risky exposure relative to what the current filtered Sharpe ratio would suggest, because a regime switch would materially change the optimal strategy. When you're confident, you invest myopically."

Transition: "Now the algorithm that learns all of this from data."

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13 Slide 13 — The POEMV Algorithm (~1:30; cumulative 13:00)

Polynomial parametrization.

$\ln f^\theta$ and g^θ are approximated by order-m polynomials in p and $(T-t)$. At $m=2$: relative mean absolute errors of 1.03% for $\ln f$ and 0.06% for g against the finite-difference truth. The log-transform of f ensures positivity. This parametrization keeps the total parameter count small — roughly 12 scalars — which is why it outperforms neural methods like DDPG/SAC: the structural form of the solution is known and exploited directly.

The parametrized policy on the slide shows ϕ_1 and ϕ_2 playing the roles of $(\rho_1-\rho_2)/\sigma$ and ρ_2/σ respectively. Convergence of the algorithm is assessed by whether $\phi_1 \to \phi_1^*$ and $\phi_2 \to \phi_2^*$.

Four-step episode loop. Walk through each step at a high level — details are in the appendix slide if needed.

1. **Filter:** Compute p_{t_k} at each step using the discretized Wonham filter and estimated parameters. This is the only step requiring pre-estimated model parameters.
2. **Critic (PE):** Gradient descent on θ minimizing the martingale loss. Any nonzero drift in $V^\theta - \Lambda \int \mathcal{H}\, ds$ along observed trajectories signals a misspecified value function. The unique minimizer is the true value function for the current policy.
3. **Actor (PG):** Score-function gradient step on ϕ. The gradient is an expectation of the score function of the policy times the TD error.
4. **Lagrange update** every 10 episodes: adjust ω toward the mean terminal wealth target.

Core point: The algorithm is the Wang-Zhou (2020) actor–critic loop with one addition: Step 1 (filtering). The loss functions and gradient formulas are structurally unchanged from the full-information case.

Likely question: "Is the martingale loss the same as TD learning?" Related but not identical. TD minimizes one-step Bellman error. The martingale loss looks at the full-trajectory gap between terminal payoff and current value estimate, accumulated over all time steps. Jia and Zhou (2022) prove its unique minimizer is the true value function, whereas TD learning requires additional conditions.

Transition: "Before seeing empirical results, it's worth confirming visually that the polynomial approximation actually captures the PDE solution well."

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14 Slide 14 — PDE Surface: $f(t, p)$ (~0:25; cumulative 13:25)

Keep this slide brief — it is visual confirmation of the polynomial approximation quality and the properties already stated for Theorem 1.

The $f(t, p)$ surface is large early in the horizon (large t-to-go) and small near T, confirming that exploration variance decays as promised. It is also larger at small p (bear market belief), reflecting the less favorable expected return environment.

This surface is produced by finite-difference solution of PDE (18) and serves as the ground truth against which the order-2 polynomial approximation achieves 1.03% relative mean absolute error.

Core point: The surface confirms visually that exploration decays over time and investment conditions are worse under bear-market belief — both properties stated in Theorem 1.

Transition: "The companion surface shows how investment intensity varies with belief."

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15 Slide 15 — PDE Surface: $\gamma^*(t, p)$ (~0:25; cumulative 13:50)

$\gamma^*(t, p)$ is the wealth-scaled investment coefficient defined as $m^*(t, x, p) = -(x - \omega)\gamma^*(t, p)$. It increases in p: higher bull-market belief means a larger position in the risky asset (when $x < \omega$). The sign flip at intermediate p values reflects the hedging demand reducing or reversing the myopic allocation.

Together, Slides 14 and 15 show that the polynomial approximation is fitting a surface with genuine structure — not a flat or trivially shaped function — which makes the 1.03% error figure meaningful rather than vacuous.

Core point: Investment intensity rises with bull-market belief and is shaped by the hedging demand, confirming the myopic-plus-hedging decomposition geometrically.

Transition: "Now let's see how this performs — starting with Wu and Li's own S&P 500 results as a reference point, before turning to our DAX application."

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16 Slide 16 — Wu & Li (2024): S&P~500 Empirical Results (~1:10; cumulative 15:00)

This slide exists to anchor the audience with the paper's core empirical finding before presenting our application. Keep it tight — we are not the authors of these results.

Setup: 30 years of S&P 500 data (09/29/1992–10/04/2022); 20-year train, 10-year test. 30 $\beta \approx 1$ stocks as index proxies, giving approximately 75,000 training episodes. EMV uses the same actor–critic architecture but ignores regime structure — this isolates the contribution of regime awareness.

What to point out in the plots:

Both panels (daily and monthly rebalancing) show POEMV consistently above EMV across the test horizon. The March 2020 COVID crash is visible as the sharp dip — POEMV shows a smaller drawdown, consistent with the Wonham filter detecting rising bear-market probability and reducing equity exposure via the hedging term.

From Table 4 of the paper: under daily rebalancing, POEMV achieves a variance of 0.224 versus EMV's 0.453 — roughly half — while both hit the mean target of 8. The Sharpe ratios are 15.38 (POEMV) versus 10.09 (EMV). Monthly rebalancing degrades both but POEMV remains superior.

Core point: Regime awareness roughly halves out-of-sample variance relative to EMV while maintaining the mean wealth target. This is the benchmark our DAX exercise is evaluated against.

Transition: "Our application takes this same algorithm and asks whether the advantage holds in a structurally different market."

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17 Slide 17 — Simulation Study: DAX Application (~0:45; cumulative 15:45)

Goal: Apply and compare POEMV and EMV on the German DAX, calibrated to 1999–2022 data, as an out-of-sample test of the regime-awareness advantage in a European setting.

Three structural differences from the S&P 500 (right column of table):

First, drift magnitudes are roughly 10× smaller. The DAX bull-market drift of 0.029 vs. 0.250 means the Sharpe ratio differences between regimes are much smaller, which makes regime detection harder — the filter has less signal to work with.

Second, the bear regime is far more persistent: average duration $\lambda_2^{-1} \approx 3$ years ($\lambda_2 = 0.339$) vs. 4 months for the S&P 500. The European sovereign debt crisis (2010–2012) falls within the training window and is a structurally different kind of bear market from the short, sharp U.S. recessions.

Third, the non-zero risk-free rate $r = 0.015$ from the ECB deposit facility enters the Sharpe ratio computations throughout. The wealth target is recalibrated to $z = 2$ (from $z = 8$) because the original target implies unreachable growth rates under DAX dynamics.

Data scarcity: Only 8 training paths survive the β-filter (vs. 30 for S&P 500), a consequence of the narrower DAX constituent universe. This is the binding practical constraint on the exercise.

Core point: The DAX parameter regime is not a minor perturbation of the S&P 500 setting — it is structurally harder, with weaker signals, more persistent bear regimes, and far fewer training paths.

Transition: "Given these constraints, here's what the training dynamics look like."

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18 Slide 18 — DAX Training Dynamics (~0:45; cumulative 16:30)

Left plot (terminal wealth): Both POEMV variants approach $z = 2$ over the course of training, with rolling-average terminal wealth stabilizing near the target by approximately 10,000 iterations. EMV plateaus below the target near 1.65 — a structural limitation. Under the DAX calibration, the unconditional average excess return is negative (the bear regime is so persistent and severe that the long-run average is negative), which means a regime-blind policy cannot reach the mean target regardless of how long it trains.

Right plot (ϕ_1 convergence): The true target is $\phi_1^* = (\rho_1 - \rho_2)/\sigma = (\mu_1 - \mu_2)/\sigma^2 \approx 8.37$. Both POEMV variants trend upward but reach only a small fraction of this target within the 20,000-iteration training budget — a direct consequence of having only 8 training paths. EMV converges to a value reflecting the unconditional (regime-blind) market calibration rather than the POEMV target.

The slide explicitly flags this: "Because we only have 8 training paths, we likely haven't achieved convergence." The upward trends are encouraging but the gap to ϕ_1^* would require substantially more training data.

Core point: The training dynamics expose a practical data requirement the theory does not yet characterize. The upward trends in ϕ_1 confirm the algorithm is working; the shortfall from ϕ_1^* confirms data scarcity is the bottleneck.

Transition: "Despite incomplete convergence, the out-of-sample results are informative."

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19 Slide 19 — DAX Out-of-Sample Results (~1:30; cumulative 18:00)

Framing the results. The slide header reads "Initial Results: quite surprising compared to Wu and Li." Both methods achieve mean terminal wealth close to 1 (initial wealth) rather than the target $z = 2$, reflecting the unconverged ϕ_1 and the inherently low-return DAX environment. This is the headline surprise: the strong S&P 500 advantage for POEMV does not replicate cleanly here.

The metrics table:

- POEMV (true): Mean 1.070, Variance 1.171, Sharpe 0.06
- POEMV (est.): Mean 1.460, Variance 0.899, Sharpe 0.49
- EMV: Mean 1.043, Variance 0.787, Sharpe 0.05
- Target z: 2.000

All three methods fall well short of $z = 2$. The mean terminal wealth of 1.070 for POEMV (true) indicates average growth of only 7% over the full 10-year out-of-sample period — consistent with the unconverged, near-zero policy under sparse training data.

The wealth plot (left): The wide bands around POEMV (true) reflect high cross-path dispersion — the regime-conditioned policy takes larger positions in bull periods and withdraws in bear periods, generating more variation across paths. The March 2020 COVID crash is visible; POEMV paths do not show a pronounced drawdown, consistent with the Wonham filter shifting toward bear-market belief and reducing equity exposure.

The counterintuitive Sharpe ordering. POEMV (est.) achieves Sharpe 0.49 versus 0.06 for POEMV (true). Wu and Li (2024) document the same pattern in their simulation study: the biased filter dampens belief-state volatility, which reduces variance in the policy gradient estimates and can stabilize parameter trajectories. With both variants far from convergence, this regularization effect likely placed POEMV (est.) in a better region of parameter space within the available iterations. It should not be interpreted as evidence that imprecise filtering is generally beneficial.

The variance ordering (EMV lowest at 0.787) should not be interpreted as better risk control. A near-inactive policy — which is what EMV becomes under DAX dynamics — mechanically produces low terminal wealth variance by taking small positions, but sacrifices the return needed to reach $z = 2$. The meaningful comparison is joint achievement of mean and variance.

Core point: The DAX results are honestly presented as surprising and incomplete. The structural challenges (sparse data, weak signal, persistent bear regime) prevent the clean replication of the S&P 500 advantage, motivating the open problem of joint estimation and control.

Transition: "These results also motivate some open theoretical questions, which brings us to the conclusion."

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20 Slide 20 — Conclusion & Open Problems (~1:00; cumulative 19:00)

The slide is organized with **Takeaways** on the left and **Open Problems** on the right. Read or closely paraphrase the takeaways first, then note the open problems.

Takeaways (left column):

1. Wonham filtering + separation principle make the martingale actor–critic viable under partial information. This is the structural insight that makes everything work.
2. Regime-awareness materially matters: POEMV roughly halves out-of-sample variance vs. EMV in the S&P 500 setting.
3. DAX results are encouraging in direction but training data scarcity is a real constraint — the unconverged policy limits what we can conclude about the European market.

Open Problems (right column): The slide presents two:

1. **Joint estimation & control:** Currently filter parameters are estimated once before training. The DAX results suggest filtering error dominates learning error as the practical bottleneck. A fully model-free architecture updating $(\tilde{\mu}_i, \tilde{\lambda}_i, \tilde{\sigma})$ alongside (θ, ϕ, ω) during training is the most impactful open problem.

2. **Scalability:** Two regimes, one risky asset. The multi-asset extension preserves the Gaussian policy structure (paper's Appendix A), but more than two regimes or continuous hidden states would require revisiting both the Wonham filter and the HJB analysis from scratch.

Two additional open problems worth having in reserve for Q&A (not on the slide): **Convergence** (a rigorous rate for the full learning loop combining filter error, approximation error, and gradient error remains open) and **time consistency** (the separation principle holds for the precommitted strategy; extending to equilibrium strategies under partial information is not known, see Dai et al. 2023 for the full-information case).

Suggested closing: "To put it concisely: Wu and Li (2024) show you can do continuous-time RL for mean-variance investing under partial information, you should, and the limiting factor in practice is how well you can estimate the regime dynamics — which is an estimation problem, not a learning one. Our DAX exercise supports that conclusion and shows where the algorithm struggles when data is scarce."

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21 General Q&A Notes

"What happens if there are more than two regimes?" The model extends to $K > 2$ regimes by replacing the scalar belief state p_t with a probability vector on K states evolving under a vector Wonham filter. The PDEs for f and g preserve the same structure but become higher-dimensional. Wu and Li note in their appendix that the multi-asset, two-regime case goes through without essential technical difficulty. More than two regimes is not addressed and would require revisiting both filter and HJB analysis.

"How sensitive is the performance to Λ?" Wu and Li use $\Lambda = 1$ throughout and note it as a tuning parameter. Higher Λ sustains more exploration (wider Gaussian policy), which can help convergence speed but increases out-of-sample variance. No systematic sensitivity analysis is provided in the paper.

"What about transaction costs?" Explicitly ignored in the paper (footnote 4). Under daily rebalancing the policy trades frequently, which would materially affect net returns under realistic transaction costs. Monthly rebalancing reduces this but also reduces performance — the performance gap between daily and monthly for the classical approach widens dramatically, which is where transaction costs matter most.

"Why is EMV variance lower than POEMV in the DAX results?" A near-inactive policy (small ϕ_1) mechanically produces low terminal wealth variance by taking small positions. EMV is essentially doing this under DAX dynamics because the unconditional average excess return is negative — the regime-blind policy has no basis for taking large positions. Low variance from inactivity is not risk control.

"Why does the estimated-filter POEMV outperform the true-filter POEMV?" Wu and Li document the same pattern in their simulation study (Table 3 of the paper) and attribute it to a regularizing effect: the biased filter dampens belief-state volatility, which reduces variance in the policy gradient estimates and can stabilize parameter trajectories. In our DAX setting — both variants far from convergence — it may additionally reflect that POEMV (est.) happened to land in a better region of parameter space within the available iterations. It should not be interpreted as evidence that imprecise filtering is beneficial in general.

"How does the leverage constraint work?" The paper applies a post-hoc truncation at execution time: if the sampled action u_{t_k} exceeds $x_{t_k} \cdot \ell$, it is clipped to $x_{t_k} \cdot \ell$. This is not learned by the algorithm. Wu and Li (2024) Table 5 shows POEMV still outperforms EMV under leverage caps of $\ell = 2$ and $\ell = 3$.

"What is the innovation process $d\hat{B}_t$ intuitively?" It is the part of the observed log-price change that is not explained by the current filtered drift estimate. Formally, $d\hat{B}_t = dS_t/(S_t\sigma) - \hat{\rho}_t\, dt$, a standard Brownian motion under $\mathcal{F}_t^S$. Think of it as the price surprise after accounting for what the filter predicts: the raw information the filter uses to update p_t.

"Isn't the precommitted strategy time-inconsistent and therefore suboptimal?" Yes, relative to a time-consistent equilibrium strategy. The precommitted strategy is optimal only at $t = 0$ and a sophisticated investor would re-optimize at each point, yielding a different (equilibrium) strategy. Dai et al. (2023) develop an RL algorithm for the equilibrium MV strategy in the full-information GBM case; extending that to partial information and regime switching requires revisiting the separation principle and is listed as an open problem.